

● RAG Learning

B5.1 Inheritance

OCR A GCSE Biology (J247) Gateway — Topic B5: Genes, inheritance and selection

1. DNA, Genes and Chromosomes

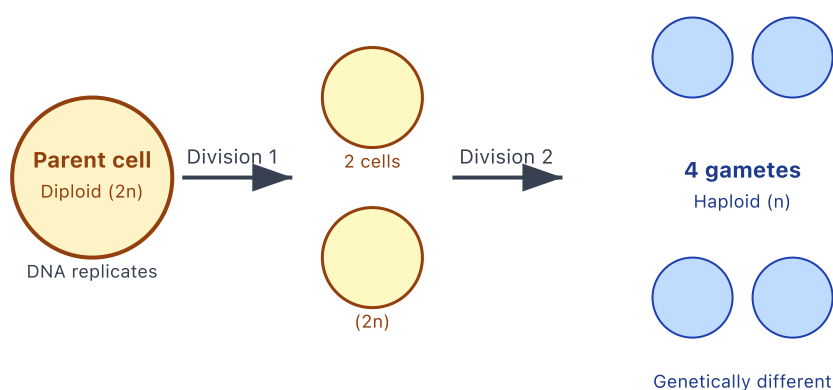
Every cell nucleus contains **chromosomes** made of long molecules of **DNA**. A short section of DNA that codes for a particular protein is a **gene**. Humans have 46 chromosomes (23 pairs) in each body cell — one of each pair from the mother and one from the father.

Term	Meaning
Genome	The entire genetic material of an organism (all the DNA)
Chromosome	Long thread of DNA found in the nucleus; contains many genes
Gene	Section of DNA coding for a particular protein
Allele	Different version of the same gene (e.g. brown vs blue eye colour)
Genotype	The combination of alleles an organism has (e.g. BB, Bb, bb)
Phenotype	The physical characteristics shown (e.g. brown eyes)
Homozygous	Both alleles the same (BB or bb)
Heterozygous	Two different alleles (Bb)
Dominant (capital letter)	Always expressed if present
Recessive (lowercase)	Only expressed when both alleles are recessive

2. Mitosis vs Meiosis

	Mitosis	Meiosis
Number of divisions	One	Two
Number of daughter cells	2	4
Genetic content	Genetically identical (clones)	Genetically different
Chromosome number	Diploid ($2n$) — same as parent	Haploid (n) — half of parent
Where	All body cells (growth, repair)	Reproductive organs (gametes)
Purpose	Growth, repair, asexual reproduction	Produce gametes (sex cells) for sexual reproduction

Meiosis — Producing 4 Genetically Different Haploid Gametes



At **fertilisation**, a male haploid gamete (sperm — 23 chromosomes) fuses with a female haploid gamete (egg — 23 chromosomes) to form a diploid zygote (46 chromosomes).

3. Sex Determination

Of the 23 pairs of human chromosomes, one pair are the **sex chromosomes**. Females are **XX** (two X chromosomes); males are **XY**. Eggs all carry an X chromosome; sperm carry either X or Y. The sex of the child is determined by whether an X-carrying or Y-carrying sperm fertilises the egg.

Punnett square for sex inheritance:

	X (egg)	X (egg)
X (sperm)	XX	XX
Y (sperm)	XY	XY

Result: 50% XX (female), 50% XY (male) — explains why male:female ratio at birth is approximately 1:1.

4. Monohybrid Inheritance — Punnett Squares

A **monohybrid cross** looks at the inheritance of a single gene with two alleles. We can predict offspring ratios using a **Punnett square**.

Worked example: Eye colour

Brown allele (B) is dominant over blue (b). Both parents are heterozygous (Bb).

	B	b
B	BB	Bb
b	Bb	bb

Genotype ratio: 1 BB : 2 Bb : 1 bb

Phenotype ratio: 3 brown eyes : 1 blue eyes

Worked example: Test cross

An organism showing the dominant phenotype (brown eyes) could be BB or Bb. Crossing with a homozygous recessive (bb) reveals which:

If parent is BB:	BB × bb → all offspring Bb (all brown)
If parent is Bb:	Bb × bb → 50% Bb (brown), 50% bb (blue)

5. Genetic Disorders

Disorder	Allele type	Symptoms
Cystic fibrosis	Recessive	Thick mucus in lungs and digestive system; difficulty breathing; recurrent infections; poor digestion
Polydactyly (extra fingers/toes)	Dominant	Extra digits at birth (usually surgically removed)
Sickle cell anaemia	Recessive (codominant heterozygous gives some protection from malaria)	Crescent-shaped red blood cells block capillaries; anaemia
Huntington's disease	Dominant	Progressive nervous system damage; usually appears in middle age
Haemophilia	X-linked recessive	Blood does not clot; mostly affects males

Worked example — cystic fibrosis

Both parents are carriers (Ff). What is the chance their child has cystic fibrosis?

	F	f
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F	FF	Ff
f	Ff	ff

Probability of cystic fibrosis (ff) = **1 in 4 = 25%**. Carrier (Ff) = 50%. Unaffected and not a carrier (FF) = 25%.

6. Embryo Screening and Genetic Engineering

Embryo screening (PGD)

In IVF, embryos can be screened for genetic disorders before being implanted. **Pros:** avoids passing on serious diseases; reduces suffering. **Cons:** ethical concerns about destroying embryos; "designer babies" slippery slope; expensive; not all conditions screenable.

Genetic engineering

Genetic engineering involves transferring genes between organisms. Examples:

- **Bacteria producing human insulin** for diabetics — gene cut from human DNA, inserted into bacterial plasmid using **restriction enzymes** and **ligase**.
- **Golden Rice** — engineered to make beta-carotene (vitamin A) to combat blindness in poor regions.
- **Pest-resistant crops** (Bt corn) — bacterial toxin gene inserted into corn.

Concerns: long-term effects unknown; possible spread to wild plants; ethics of patenting living things; reduced biodiversity if monocultures dominate.

7. Variation

Source of variation	Examples

Genetic	Eye colour, blood group, hair texture (passed down from parents via gametes)
Environmental	Tan from sunlight; scars; weight from diet/exercise
Combination	Most traits — e.g. height influenced by genes AND nutrition

Mutation is a random change in the DNA sequence. Most mutations have no effect (silent), some are harmful, and rarely some are beneficial. Beneficial mutations are the raw material for evolution by natural selection.